GlucoTwin

A digital twin for Type 2 Diabetes that predicts glucose spikes 2 hours ahead, explains why, and simulates what would prevent them

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Happiest Health Digital Twin Challenge 2026

All data synthetic · research prototype, not a medical device

Between clinic visits, nobody sees the spike coming

100M+

people in India live with diabetes, most with type 2

90 days

between HbA1c tests: one number for a whole quarter

288

CGM readings a day that sit apart from the health record

The data exists, but it is split in two

The EHR explains who the patient is. Wearables show what their glucose is doing. Clinicians get either a lab value every three months or a raw CGM trace.

Neither tells them what is about to happen, or what would change it.

One twin per patient: predict, explain, simulate

1

Predict

Probability that glucose will exceed 180 mg/dL in the next 2 hours, updated every 5 minutes

2

Explain

Top 3 reasons in plain English from SHAP, e.g. "Ate 105 g carbs, 10 min ago — raises risk"

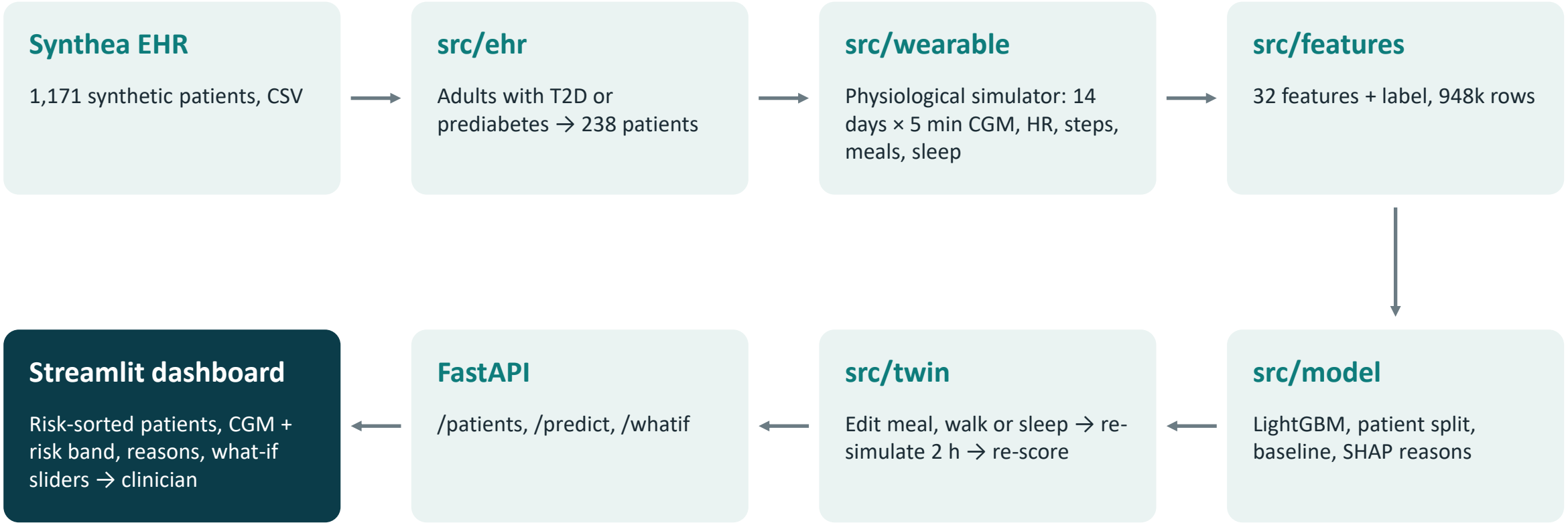
3

Simulate

What-if: a smaller meal, a 20-minute post-meal walk or a better night's sleep, as a new glucose curve and risk

Inputs fused per patient: static EHR (age, sex, BMI, HbA1c, diagnoses, metformin, lipids) + dynamic wearables (CGM every 5 min, steps, heart rate, sleep, meals)

From synthetic records to a doctor's dashboard



The twin reuses the same simulator and feature code as training, so what-if results are scored by the same model.

Wearable simulator grounded in diabetes physiology

A1c

Baseline from HbA1c

eAG (mg/dL) = 28.7 × HbA1c − 46.7. Simulated mean tracks eAG (r = 0.945)

M

Indian carb-heavy meals

3 meals + optional snack. Peak 45–75 min, scaled by carbs × insulin resistance (BMI, HbA1c)

S

Steps lower glucose

Effect lasts 1–3 h, strongest after meals. Heart rate rises with activity

Z

Poor sleep raises resistance

Under 6 h or low efficiency → higher next-day insulin resistance

Rx

Metformin

Blunts the meal rise by 15%

~

Realistic noise

Physiological drift + CGM sensor noise, clipped to 40–400 mg/dL. Deterministic per seed

LightGBM on 32 fused features, split by patient

Task	Will max glucose in (t, t+120 min] exceed 180 mg/dL?
Unit	(patient, 5-minute timestamp)
Features	Glucose now, 15/30/60-min lags, rate of change, rolling stats, carbs and steps in last 1–2 h, minutes since meal, heart rate, last night's sleep, time of day, static EHR
Model	LightGBM · 500 trees · learning rate 0.05 · 31 leaves
Baseline	Naive rule: current glucose > threshold (swept)
Explain	SHAP TreeExplainer, related features grouped into plain-English reasons

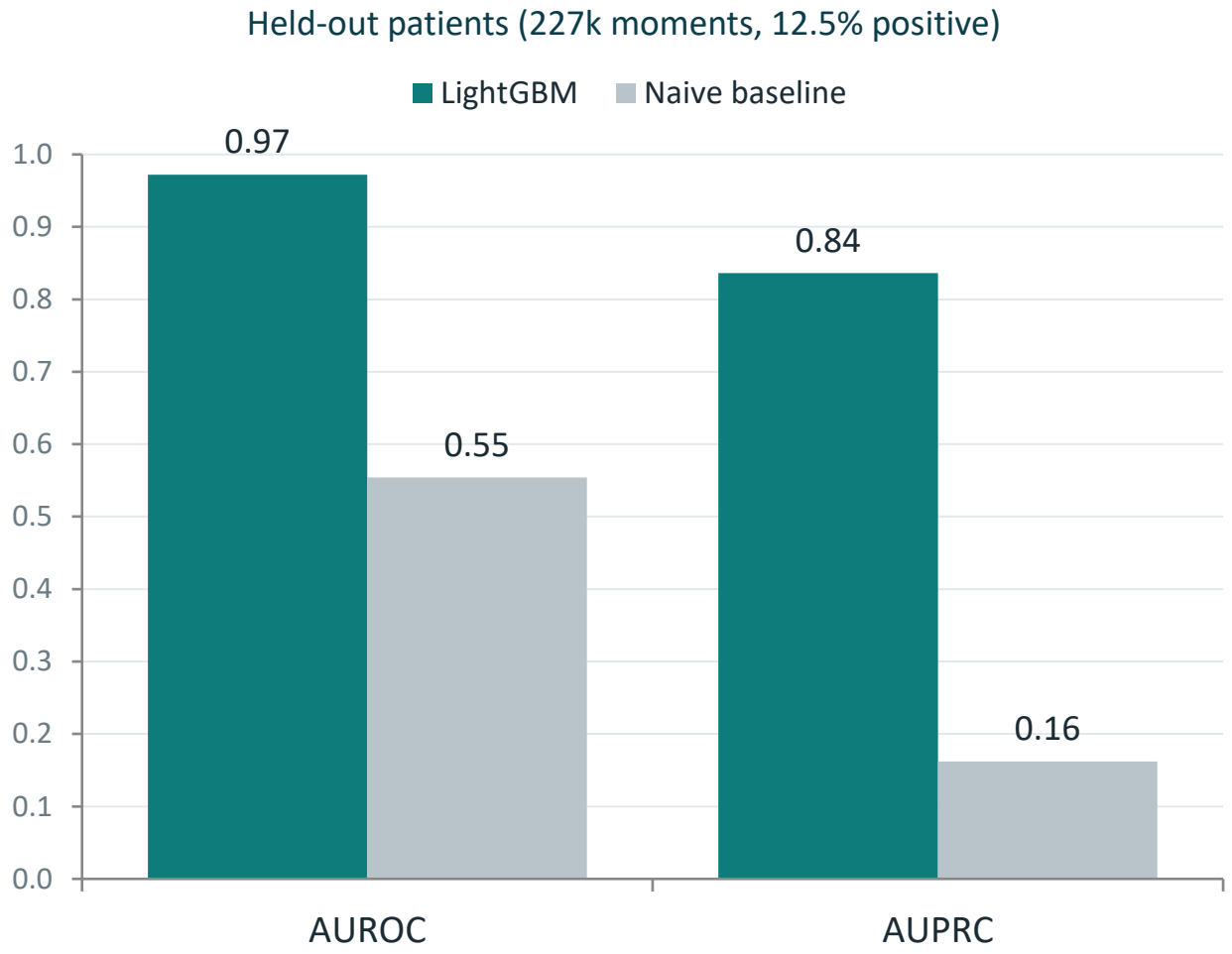
No leakage between patients

178 train patients

60 test patients

GroupShuffleSplit by patient, never by row, checked by a unit test. Evaluated on 227,025 test moments where glucose is still ≤ 180, so the model must predict an upcoming spike.

Unseen patients: far better than the naive rule



95%

of spike events alerted in advance (1,219 / 1,282)

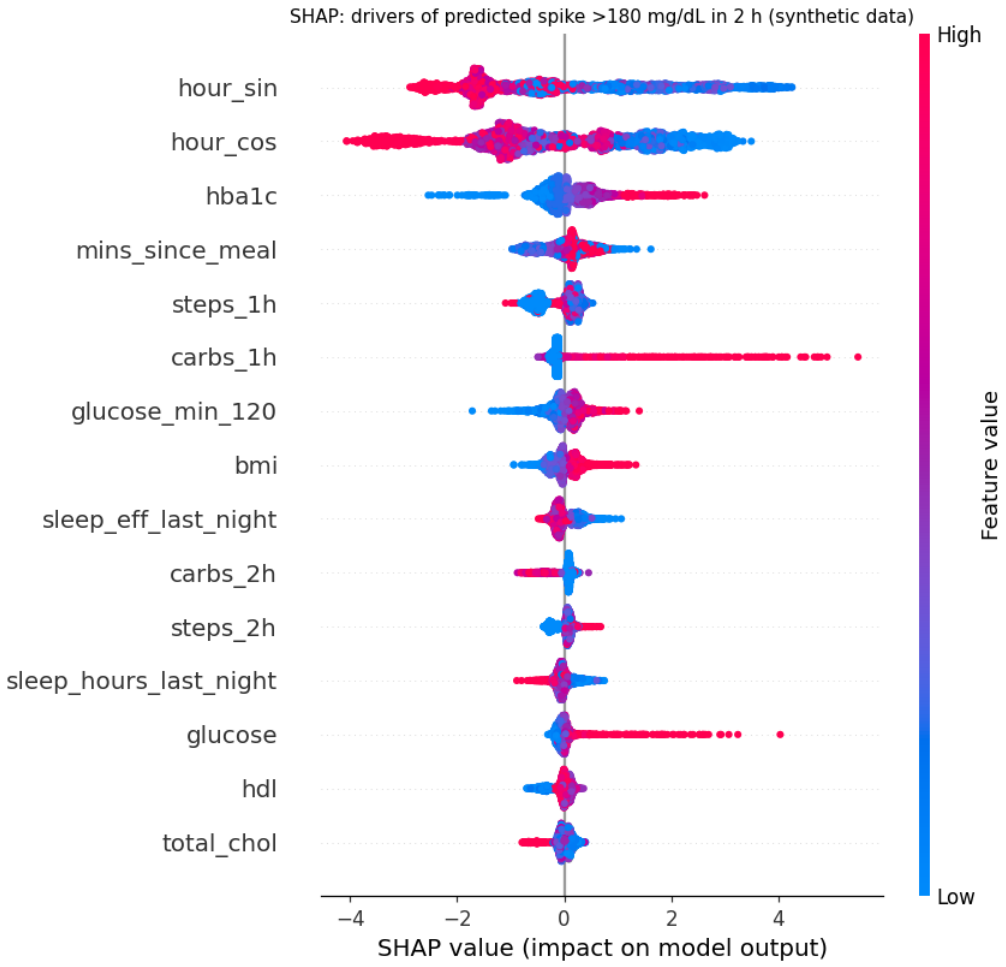
95 min

median warning before glucose crosses 180

0.66

recall at 80% precision (baseline: 0.00)

Every alert comes with reasons a clinician can check



Example reason shown on the dashboard

"Ate 105 g carbs, 10 min ago — raises risk"

Directions match physiology

- More carbs, higher HbA1c and higher BMI raise risk
- Recent walking and better sleep lower it
- Top drivers: time of day, HbA1c, minutes since meal, steps and carbs in the last hour
- Related SHAP features are grouped into the top 3 reasons per prediction

A walk or a smaller lunch prevents the spike

Demo patient be33d994

51-year-old woman with T2D, on metformin, BMI 29.3, hypertension

Slept 5.2 h, then ate a 105 g rice-heavy lunch at 13:15

Model risk at 13:25

98%

Twin re-simulation of next 2 h	Spike prob.	Peak	Min > 180
As usual	**100%**	**198 mg/dL**	**40**
20-minute post-meal walk	36%	173 mg/dL	**0**
60 g lunch instead of 105 g	**0%**	160 mg/dL	**0**
Walk and 60 g lunch	**0%**	**137 mg/dL**	**0**

Across all 238 patients' lunches, a 20-minute walk lowers the simulated peak by **24 mg/dL** on average and spike probability from **74% to 33%**

Dashboard: CGM, twin projection and what-if



Blue: observed CGM · **Dotted:** twin projection as usual (peaks at 198) · **Green:** what-if with walk + smaller lunch · **Red band:** predicted risk

Healthcare use case: triage now, coach with evidence

1

Rank

Dashboard sorts all patients by current spike risk, so the care team sees who needs a nudge now

2

Understand

EHR card, 24 h CGM with risk band and the top 3 reasons for this patient

3

Test

Move what-if sliders for meal carbs, a post-meal walk or sleep and see the new curve

4

Coach

Show the patient which change helps them most, using their own twin

Who benefits: diabetologists and diabetes educators managing many T2D patients between visits, and patients who get personalised, evidence-backed advice instead of generic rules.

What these results do and do not show

The model partly learns our simulator

Metrics show how well it recovers our physiology rules, not real-world accuracy. Validation on real CGM data is future work.

Time of day is too strong

Everyone eats at similar times, so the clock alone reaches AUROC 0.935. Without time features: AUROC 0.927, AUPRC 0.68.

Population mismatch

Synthea models a US population; HbA1c tops out at 7.5% (39 T2D, 199 prediabetes). Meals are Indian, demographics are not.

Single-lever what-ifs

Meal composition (fibre, protein, GI), medication timing and stress are not modelled.

All data is synthetic (Synthea + our simulator): no real person's data, safe under DPDP Act 2023 and HIPAA. GlucoTwin is a research prototype, not a medical device.

Next steps

From a convincing prototype to a validated twin

- Validate on real T2D CGM datasets
- Personal meal schedules to tame the time-of-day signal
- Meal composition, medication timing and stress in the simulator
- Consent, encryption, access control and audit logging
- Clinician pilot of the dashboard

Thank you

Team GlucoTwin · BIT Mesra

github.com/swinter-ctrl/GlucoTwin_BITMesra

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